

Oracle DBA Cheat Sheet

Topic	Details	Check Query/Command
Oracle Database	Robust RDBMS for high-performance data management.	SELECT * FROM v\$version;
RMAN Backup	Consistent (Cold, offline, MOUNT mode), Inconsistent (Hot, online, ARCHIVELOG mode), Full (all data), Incremental (daily changes).	RMAN> report schema; RMAN> list backup;
RMAN Cold Backup	SHUTDOWN IMMEDIATE; STARTUP MOUNT; BACKUP DATABASE;	SELECT status FROM v\$instance;
RMAN Hot Backup	BACKUP DATABASE PLUS ARCHIVELOG; (Requires ARCHIVELOG mode).	ARCHIVE LOG LIST;
Data Guard	Standby: Clone of production DB. Types: Physical (block-by-block, same DB ID), Logical (SQL apply, separate DB ID), Snapshot (testing with flashback). Processes: LNS (redo transport), MRP (physical apply), LSP (logical apply). Modes: Max Performance, Max Availability, Max Protection.	SELECT protection_mode FROM v\$database; SELECT database_role FROM v\$database; SELECT switchover_status FROM v\$database; SELECT process, status FROM v\$managed_standby;
Data Guard Parameters	DB_NAME, DB_UNIQUE_NAME, LOG_ARCHIVE_DEST_n, FAL_SERVER, STANDBY_FILE_MANAGEMENT, LOG_ARCHIVE_CONFIG.	SHOW PARAMETER db_name; SHOW PARAMETER db_unique_name; SHOW PARAMETER log_archive_dest; SHOW PARAMETER fal_server; SHOW PARAMETER standby_file_management; SHOW PARAMETER log_archive_config;
RAC	One database, multiple instances for scalability.	SELECT inst_id, instance_name, status FROM gv\$instance;
ASM	Automatic Storage Management for simplified storage.	SELECT name, state FROM v\$asm_diskgroup;
GoldenGate	Real-time data replication using capture/apply.	SELECT capture_name, status FROM dba_capture; SELECT apply_name, status FROM dba_apply;
Oracle Architecture	Instance: SGA (Database Buffer Cache: stores data blocks; Redo Log Buffer: holds redo entries; Shared Pool: SQL, PL/SQL, data dictionary, library cache; Large Pool: RMAN, parallel execution; Java Pool: Java execution), PGA (session memory for sorting), Background Processes (DBWn: writes buffers; LGWR: writes redo logs; CKPT: updates headers; SMON: recovery; PMON: cleans failed processes). Database: Datafiles, control files, redo logs.	SELECT name, value FROM v\$sga; SELECT component, current_size/1024/1024 MB FROM v\$sga_dynamic_components; SELECT process FROM v\$bgprocess WHERE paddr != '00'; SELECT name, value/1024/1024 MB FROM v\$pgastat WHERE name IN ('aggregate PGA target parameter', 'total PGA allocated');
User Process	Client process initiating SQL requests.	SELECT sid, username, program FROM v\$session WHERE type='USER';

Server Process	Executes SQL for user.	SELECT sid, spid, program FROM v\$process WHERE addr IN (SELECT paddr FROM v\$session);
Physical Files	Datafiles (.dbf), control files (.ctl), redo logs (.log), archive logs, spfile/pfile.	SELECT name FROM v\$datafile; SELECT name FROM v\$controlfile; SELECT member FROM v\$logfile;
Logical Files	Tablespaces (8KB default block size), segments, ex- tents, blocks.	SELECT tablespace_name, segment_name FROM dba_segments;
File Paths	Datafiles: /u01/oradata/DBNAME/, Control: /u01/control/, Redo: /u01/redolog/.	SELECT file_name FROM dba_data_files; SELECT name FROM v\$controlfile; SELECT member FROM v\$logfile;
TNS	Transparent Network Substrate (TNSNAMES.ORA).	tnsping <service_name>
DB Link	Access remote database objects.	SELECT db_link, host FROM dba_db_links;
Patching	Resolves bugs and vulnerabilities.	opatch lsinventory;
Hot Backup	ALTER TABLESPACE tbs BEGIN BACKUP; copy files; ALTER TABLESPACE tbs END BACKUP;.	SELECT tablespace_name, status FROM dba_- tablespaces WHERE status='BACKUP';
EXPDP (Full Backup)	expdp system/pwd FULL=Y DIRECTORY=dir DUMPFILE=full_exp.dmp LOGFILE=full_exp.log	SELECT job_name, operation, state FROM dba_- datapump_jobs WHERE operation='EXPORT';
EXPDP (Schema/Table)	expdp system/pwd SCHEMAS=schema_- name DIRECTORY=dir DUMPFILE=schema.dmp LOGFILE=schema.log	SELECT job_name, operation, state FROM dba_- datapump_jobs WHERE operation='EXPORT';
IMPDP	impdp system/pwd DIRECTORY=dir DUMPFILE=full_- exp.dmp LOGFILE=full_imp.log	SELECT job_name, operation, state FROM dba_- datapump_jobs WHERE operation='IMPORT';
Data Pump Tuning	Use DIRECT=Y, PARALLEL=n, schedule via crontab.	SELECT job_name, degree FROM dba_datapump_- jobs;
Data Pump Directory	OS: mkdir /path/dir, DB: CREATE DIRECTORY dir AS '/path/dir';	SELECT directory_name, directory_path FROM dba_directories;
Network Import	impdp system/pwd NETWORK_LINK=dblink DUMPFILE=exp.dmp LOGFILE=imp.log	SELECT db_link, host FROM dba_db_links;
Grant Privilege	GRANT SELECT ON table TO user;	SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE grantee='USER';
Revoke Privilege	REVOKE SELECT ON table FROM user;	SELECT grantee, privilege, table_name FROM dba_tab_privs WHERE grantee='USER';
User Management	CREATE USER user IDENTIFIED BY pwd; ALTER USER user IDENTIFIED BY new_pwd; DROP USER user;	SELECT username, account_status FROM dba_- users;
Role Management	CREATE ROLE role_name; GRANT SELECT ON table TO role_name;	SELECT role, privilege FROM dba_role_privs WHERE grantee='ROLE_NAME';

Profile Management	CREATE PROFILE prof LIMIT PASSWORD_LIFE_TIME 180;	SELECT profile, resource_name, limit FROM dba_profiles WHERE profile='PROF';
Add Datafile	ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M;	SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE tablespace_name='TBS';
Resize Datafile	ALTER DATABASE DATAFILE '/path/file.dbf' RESIZE 200M;	SELECT file_name, bytes/1024/1024 MB FROM dba_data_files WHERE file_name='/path/file.dbf';
Check Tablespace Size	SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;	SELECT tablespace_name, SUM(bytes)/1024/1024 MB FROM dba_data_files GROUP BY tablespace_name;
Alter Tablespace	ALTER TABLESPACE tbs ADD DATAFILE '/path/file.dbf' SIZE 100M AUTOEXTEND ON;	SELECT tablespace_name, file_name, autoextensible FROM dba_data_files WHERE tablespace_name='TBS';
SQL Utility	Clear SQL terminal: cl scr	cl scr
DBA Handover Checks	Check alert log, backups, tablespace space, listener, RAC health, Data Guard sync, job schedules.	SELECT * FROM v\$alert_log; SELECT destination, status FROM v\$archive_dest; SELECT tablespace_name, (SUM(bytes)/1024/1024) - (SUM(free_space)/1024/1024) MB_USED FROM dba_data_files JOIN dba_free_space USING(tablespace_name) GROUP BY tablespace_name; lsnrctl status; SELECT inst_id, status FROM gv\$instance; SELECT sequence#, applied FROM v\$archived_log; SELECT job_name, state FROM dba_scheduler_jobs;

Oracle Files and Paths

Oracle Files Explained

File Type	Extension	Details	Analogy
Data File	.dbf	Stores tables, indexes, temporary/undo blocks, materialized views, LOBs.	Notebook with actual data.
Redo Log File	.log	Sequential log of DML changes (insert/update/delete), transaction vectors.	CCTV live recording.
Archived Log	.arc	Backup of redo logs for media/point-in-time recovery.	CCTV backup hard drive.
Control File	.ctl	DB structure, SCN, datafile names, redo log sequence, backup info.	Books table of contents.
Parameter File	spfile/init.ora	DB startup config: DB_NAME, memory sizes, paths.	Cars ignition settings.
Trace File	.trc	Errors, diagnostics, session debug info.	Doctors patient report.
Alert Log	.log	DB events: startup/shutdown, errors, tablespace issues.	Hospital daily log.

Summary Table

File	Data Type	Written By
.dbf	Table data	DBWR
.log (redo)	Transaction logs	LGWR
.arc	Redo log backup	ARCH
.ctl	Metadata & structure	Oracle engine
spfile/init.ora	Startup parameters	DBA or Oracle
.trc	Debug info	Oracle processes
alert.log	DB-wide event log	Oracle instance

Oracle DBA Important Paths

Purpose	Path / File Location	Description
Oracle Base	\$ORACLE_BASE (/u01/app/oracle)	Root folder for Oracle software, logs, configs.
Oracle Home	\$ORACLE_HOME (/u01/app/oracle/product/19.0.0/db_1)	Oracle software installation location.

Listener Config	/u01/app/oracle/product/19.0.0/db_1/network/admin/listener.ora	Listener port, host configuration.
TNS Config	/u01/app/oracle/product/19.0.0/db_1/network/admin/samples/listener.ora	Connection strings to DBs.
SQL*Net Config	/u01/app/oracle/product/19.0.0/db_1/network/admin/tnsnames.ora	Network protocols, encryption settings.
AWR Report Script	/u01/app/oracle/product/19.0.0/db_1/network/admin/samples/tnsnames.ora	Script to generate AWR reports for performance analysis.
Listener Log	/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/awrrpt.sql	Listener connection attempts and errors.
Listener Trace	/u01/app/oracle/diag/tnlsnr/<hostname>/listener/alert/<inst>.log	Listener connection attempts and errors.
DB Alert Log	/u01/app/oracle/diag/tnlsnr/<hostname>/listener/trace/<inst>.log	DB errors, startup/shutdown, events.
Datafiles	SELECT name FROM v\$datafile; (e.g., /u01/oradata/DBNAME/system01.dbf, users01.dbf)	Actual data file paths for tablespaces.
Redo Log Files	SELECT member FROM v\$logfile; (e.g., /u01/redolog/DBNAME/redo01.log)	Redo log file paths for crash recovery.
Control Files	SELECT name FROM v\$controlfile; (e.g., /u01/control/DBNAME/control01.ctl)	Control file paths for DB structure.
Archive Log Files	/u01/app/oracle/fast_recovery_area/<db_name>/archivelog/	Archived redo logs for recovery.
Audit Logs	\$ORACLE_BASE/admin/<db_name>/adump/	User activity logs.
RMAN Backup	/u01/app/oracle/backup/ or FRA	DB backups via RMAN.
Background Trace	/u01/app/oracle/diag/rdbms/<db>/<inst>/trace/	Trace files for SMON, PMON, DBWn.
ORATAB File	/etc/oratab	Lists all DB instances on machine.
Password File	\$ORACLE_HOME/dbs/orapw<ORACLE_SID>	SYSDBA remote login.
Logical Files	SELECT tablespace_name, segment_name FROM dba_segments; (e.g., Tablespaces: SYSTEM, SYSAUX, USERS)	Logical structures: tablespaces, segments, extents.

Oracle DBA Responsibilities and Utilities

Linux Commands for Oracle DBA

Command	Purpose
<code>netstat -tulnp</code>	Show listening ports with process info (TCP + UDP).
<code>ss -tulnp</code>	Faster alternative to <code>netstat</code> .
<code>lsof -i :1521</code>	Check process using port 1521.
<code>ps -ef grep tnslnsr</code>	Check if Oracle listener is running.
<code>lsnrctl status</code>	Show listener status, port, services.
<code>systemctl status firewalld</code>	Check firewall status.
<code>firewall-cmd --list-all</code>	Show open ports in firewalld.
<code>iptables -L -n -v</code>	List iptables rules.
<code>hostname -I</code> or <code>ip a</code>	Check system IP address.
<code>curl -v</code>	Test port 1521 connectivity.
<code>telnet://localhost:1521</code>	
<code>telnet localhost 1521</code>	Check if port accepts connections.
<code>nc -zv 127.0.0.1 1521</code>	Check port status using netcat.
<code>whoami</code>	Verify current user (e.g., oracle).
<code>find \$ORACLE_HOME -name 'tnsnames.ora'</code>	Locate <code>tnsnames.ora</code> files.
<code>find \$ORACLE_HOME -name 'listener.ora'</code>	Locate <code>listener.ora</code> files.
<code>find \$ORACLE_HOME -name 'awrrpt.sql'</code>	Locate <code>awrrpt.sql</code> script.

DBA Responsibilities

Task	Details	Commands
DB Installation	Install Oracle software, create DB, configure listener.	<code>./runInstaller -silent -responseFile db_install.rsp</code> <code>dbca -silent -createDatabase -gdbName ORCL</code> <code>lsnrctl start</code>
User Management	Create users, assign roles/privileges, manage profiles.	<code>CREATE USER app_user IDENTIFIED BY pwd;</code> <code>GRANT SELECT ON table TO app_user;</code> <code>CREATE PROFILE app_profile LIMIT PASSWORD_LIFE_TIME 90;</code>
Performance Tuning	Optimize queries with indexes, stats, memory tuning.	<code>EXPLAIN PLAN FOR SELECT ...;</code> <code>CREATE INDEX idx ON table(col);</code> <code>EXEC DBMS_STATS.GATHER_TABLE_STATS('SCOTT', 'TABLE');</code>

Backup & Recovery	Full/incremental backups, restore/recover data.	<pre> RMAN> BACKUP DATABASE PLUS ARCHIVELOG; RMAN> RESTORE DATAFILE '/path'; RMAN> RECOVER DATABASE; </pre>
Security Management	Strong passwords, auditing, TDE for encryption.	<pre> ALTER SYSTEM SET ENCRYPTION KEY IDENTIFIED BY "pwd"; AUDIT SELECT ON table BY ACCESS; ALTER USER user ACCOUNT LOCK; </pre>
Monitoring	Check alert log, CPU, memory, disk space, AWR reports.	<pre> tail -f alert_ORCL.log SELECT name, value FROM v\$sysstat WHERE name LIKE '%CPU%'; BEGIN DBMS_WORKLOAD_REPOSITORY.CREATE_- SNAPSHOT(); END; </pre>

Startup & Shutdown

Command	Description	Use Case
<pre> STARTUP; STARTUP NOMOUNT; STARTUP MOUNT; STARTUP FORCE; SHUTDOWN NORMAL; SHUTDOWN IMMEDIATE; SHUTDOWN ABORT; lsnrctl status/start/stop SELECT INSTANCE_- NAME, STATUS FROM V\$INSTANCE; </pre>	Full DB start (Instance + Mount + Open). Start instance only. Mount DB without opening. Force restart DB. Wait for users to log off. Disconnect users immediately. Force shutdown. Manage listener. Check DB status.	Normal startup. Create DB, restore control files. Media recovery, rename datafiles. Recover from issues. Graceful shutdown. Safe shutdown. Last resort. Check or control listener. Verify DB state.

Alert Log Checks

- **Startup/Shutdown:** `tail -f /u01/app/oracle/diag/rdbms/oradb/oradb/alert/alert_oradb.log`
- **Listener Issues:** Check for ORA-12541: TNS:no listener.
- **Errors:** Look for ORA-00600, ORA-01578, etc.
- **Tablespace Issues:** Check ORA-01653 for space errors.
- **Archiver Status:** Verify ARC0: Completed archiving.
- **Checkpoint/Recovery:** Look for Media Recovery Start.
- **Background Failures:** Check for SMON, DBWO errors.

- **Parameter Changes:** Look for ALTER SYSTEM logs.
- **PDB Status:** Check Pluggable database PDB1 opened.

Performance Tuning and Day-to-Day Queries

Index Rebuilding & Tuning

- Identify problematic SQL using AWR reports or V\$ views (e.g., V\$SQL, V\$SESSION).
- Check for full table scans via EXPLAIN PLAN.
- Optimize with indexes, bind variables, or query rewriting.
- Monitor hard parsing (high CPU usage).

Commands:

- `CREATE INDEX idx_salary ON employee(salary);`
- `EXPLAIN PLAN FOR SELECT * FROM employee WHERE salary > 5000;`
- `SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY);`
- `SET AUTOTRACE ON;`

AWR Report Usage

- Analyze top 5 foreground processes, CPU utilization, disk I/O, parsing (soft vs. hard), redo generation, transactions/sec.
- Ensure Buffer Cache Hit Ratio, Library Cache Hit Ratio, Soft Parse Ratio are 90100%.

Command:

- `@/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/awrrpt.sql`

SQL Performance Check

- Use EXPLAIN PLAN to identify full table scans or missing indexes.
- Suggest query optimization to developers.

Commands:

- `EXPLAIN PLAN FOR <query>;`
- `SELECT * FROM TABLE(DBMS_XPLAN.DISPLAY);`
- `SET AUTOTRACE ON;`

SQL Developer Connection Issues

- Check `listener.ora`, `tnsnames.ora` for port/service name.
- Verify listener status.

Commands:

- `lsnrctl status`
- `tnsping service_name`

Real-Life Example

- *Scenario:* System slowed down, users complained. Used AWR report to identify top processes, found full table scan via V\$SESSION and EXPLAIN PLAN. Suggested query rewrite or index creation, improving performance.

Load Profile Monitoring

- Monitor redo generated, logical/physical reads, DB time, transactions/sec.
- High hard parsing: Use bind variables.
- High transactions/sec: Check commit/rollback usage.

Day-to-Day Queries

- **Top SQLs by CPU Usage:**

```
SELECT *
FROM (
    SELECT sql_text, sql_id, executions, cpu_time, elapsed_time
    FROM v$sql
    WHERE executions > 0
    ORDER BY cpu_time DESC
)
WHERE ROWNUM <= 10;
```

- **Active Sessions:**

```
SELECT sid, serial#, username, status, osuser, machine, program, sql_id
FROM v$session
WHERE status = 'ACTIVE'
AND username IS NOT NULL;
```

- **Background/Foreground Processes:**

```
SELECT p.pid, p.spid, s.sid, s.serial#, s.username, s.program
FROM v$process p
JOIN v$session s ON p.addr = s.paddr
WHERE s.username IS NOT NULL;
```

- **Wait Events:**

```
SELECT event, total_waits, time_waited, average_wait
FROM v$system_event
WHERE event NOT LIKE 'SQL*Net message%'
ORDER BY time_waited DESC;
```

- **Top SQL by Elapsed Time:**

```
SELECT sql_id, sql_text, elapsed_time, executions
FROM v$sql
WHERE executions > 0
ORDER BY elapsed_time DESC
FETCH FIRST 5 ROWS ONLY;
```

- **Session Waits:**

```
SELECT sid, event, wait_class, seconds_in_wait, state
FROM v$session_wait
WHERE wait_class != 'Idle';
```

SQL*Plus Format Commands

```
SET LINESIZE 200;
SET PAGESIZE 100;
SET TRIMSPPOOL ON;
SET FEEDBACK OFF;
COLUMN sid FORMAT 9999;
COLUMN serial# FORMAT 99999;
COLUMN username FORMAT A15;
COLUMN sql_id FORMAT A13;
COLUMN event FORMAT A40;
COLUMN sql_text FORMAT A60;
COLUMN program FORMAT A30;
```

Interview Tips

- Explain issues confidently: e.g., I used AWR to find full table scans, suggested indexing, and improved performance.
- Use V\$ views (V\$SQL, V\$SESSION, V\$PROCESS, V\$EVENT) for monitoring.
- Collaborate with developers for query optimization.
- Practice commands and real-life scenarios.

Important Oracle Reports

Important Oracle Reports

Report	Generation Command/Path	What to Analyze	Key Points to Check
AWR Report	@/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/awrrpt.sql	System-wide performance: CPU usage, I/O, wait events, top SQLs, memory usage.	• Load Profile: Transactions/sec, redo generation, logical/physical reads. • Top 5 Foreground Events: Identify bottlenecks (e.g., db file sequential read, latch free). • Cache Hit Ratios: Buffer Cache (>90%), Library Cache (>95%), Soft Parse Ratio (>90%). • Top SQLs: High CPU/elapsed time SQLs, full table scans. • Instance Efficiency: Check for high hard parsing.
ADDM Report	@/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/addmrpt.sql	Performance issues and recommendations: SQL tuning, memory allocation, I/O bottlenecks.	• Findings: Identify high-impact issues (e.g., poorly tuned SQL, undersized SGA). • Recommendations: Apply suggested indexes, adjust memory parameters. • Bottlenecks: Check for I/O or CPU constraints. • Wait Events: Focus on top wait events impacting performance.
ASH Report	@/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/ashrpt.sql	Active session history: Real-time session activity, wait events, SQL execution.	• Top Sessions: Identify sessions consuming CPU or causing waits. • Wait Classes: Check for I/O, concurrency, or application waits. • Top SQLs: Pinpoint slow or resource-intensive queries. • Blocking Sessions: Detect locks or contention.

Statspack Report	@/u01/app/oracle/product/19.0.0/db_1/rdbms/admin/spreport.sql	Historical performance data (alternative to AWR for non-licensed systems).	• Load Profile: Transactions, redo, logical reads. • Top Wait Events: Identify bottlenecks (e.g., log file sync, db file read). • Buffer Cache: Check hit ratios and contention. • Top SQLs: Focus on high CPU/elapsed time queries.
SQL Tuning Advisor Report	<pre>EXEC DBMS_SQLTUNE.CREATE_TUNING_TASK(sql_id => '<sql_id>'); EXEC DBMS_SQLTUNE.EXECUTE_TUNING_TASK(task_name => '<task_name>'); SELECT DBMS_SQLTUNE.REPORT_TUNING_TASK('<task_name>') FROM DUAL;</pre>	Specific SQL performance: Execution plan, index usage, statistics.	• Execution Plan: Check for full table scans or costly operations. • Recommendations: Create indexes, gather stats, or rewrite SQL. • Statistics: Ensure table/index stats are up-to-date. • Benefit Analysis: Evaluate performance improvement potential.